

F1 Race Strategy Advisor

Conditional strategy choice depends on the target: the one-stop M-H plan protects Top 10 probability, while the two-stop M-H-S plan improves Top 5 upside in the approved Dutch GP scenario.

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Repo: <https://github.com/Gottch7897/iit414w-lab01-TheUltrakills.git>

Commit: final-v1

1. Executive Summary

The advisor supports a race strategy engineer making a pre-race call after qualifying. It compares one-stop and two-stop plans in a fixed driver-race context, reporting calibrated probabilities for points security (`is_top10`) and stronger upside (`is_top5`). In the Dutch GP scenario, the one-stop protects Top 10 probability, while the two-stop improves Top 5 probability. The recommendation is conditional, not automatic.

The locked `is_top10` model has Brier 0.139039 and ROC-AUC 0.881266, close to but below the decent reference of Brier 0.132 and ROC-AUC 0.892. The report therefore treats the model as useful decision support, not as a deployed strategy oracle.

2. Problem Framing

The supported decision is a pre-race strategy call before lights out, after qualifying information is available. The decision-maker is a race strategy engineer comparing whether a selected driver-race context should prioritize a one-stop or two-stop plan. The prediction unit is one driver in one race.

The primary target is `is_top10`; the expansion target is `is_top5`. Strategy fields are scenario inputs, not ordinary pre-race predictors. They would be leakage if used as observed post-race facts, but here they are user-controlled what-if values.

3. Data and Validation

The race-level dataset has 2,447 driver-race rows from 2019-2024. The locked temporal split is train 2019-2021, calibration 2022, and final test 2023-2024. The feature audit uses pre-race context plus scenario inputs and excludes outcomes, realized weather, race incidents, DNF, and status columns from prediction.

Known limitations include short historical coverage, empty qualifying_time_s, coarse safety-car indicators, and strategy choice confounding with car pace, driver quality, traffic, weather, and race incidents.

4. Modeling Approach

The final narrative uses separate logistic-regression models per target. Numeric features are imputed and standardized; categorical features are imputed and one-hot encoded. Both targets use sigmoid/Platt calibration on 2022, selected because the calibration block is small and a smooth calibrator is less likely to overfit than isotonic calibration.

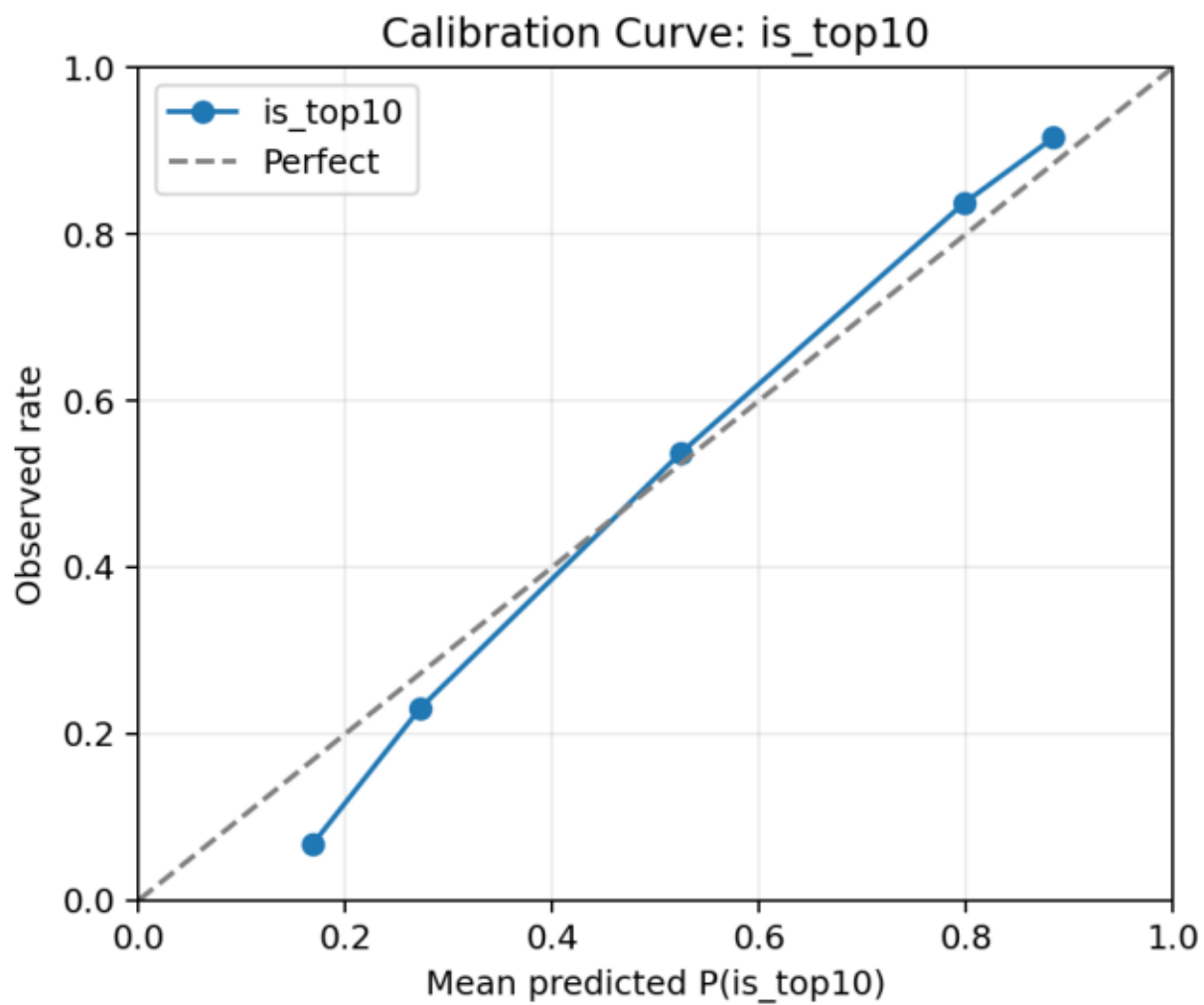
The model family is conservative and explainable. The cost is possible underfitting, which helps explain why the Top 10 model is close to but below the docent reference.

5. Results and Honest Comparison

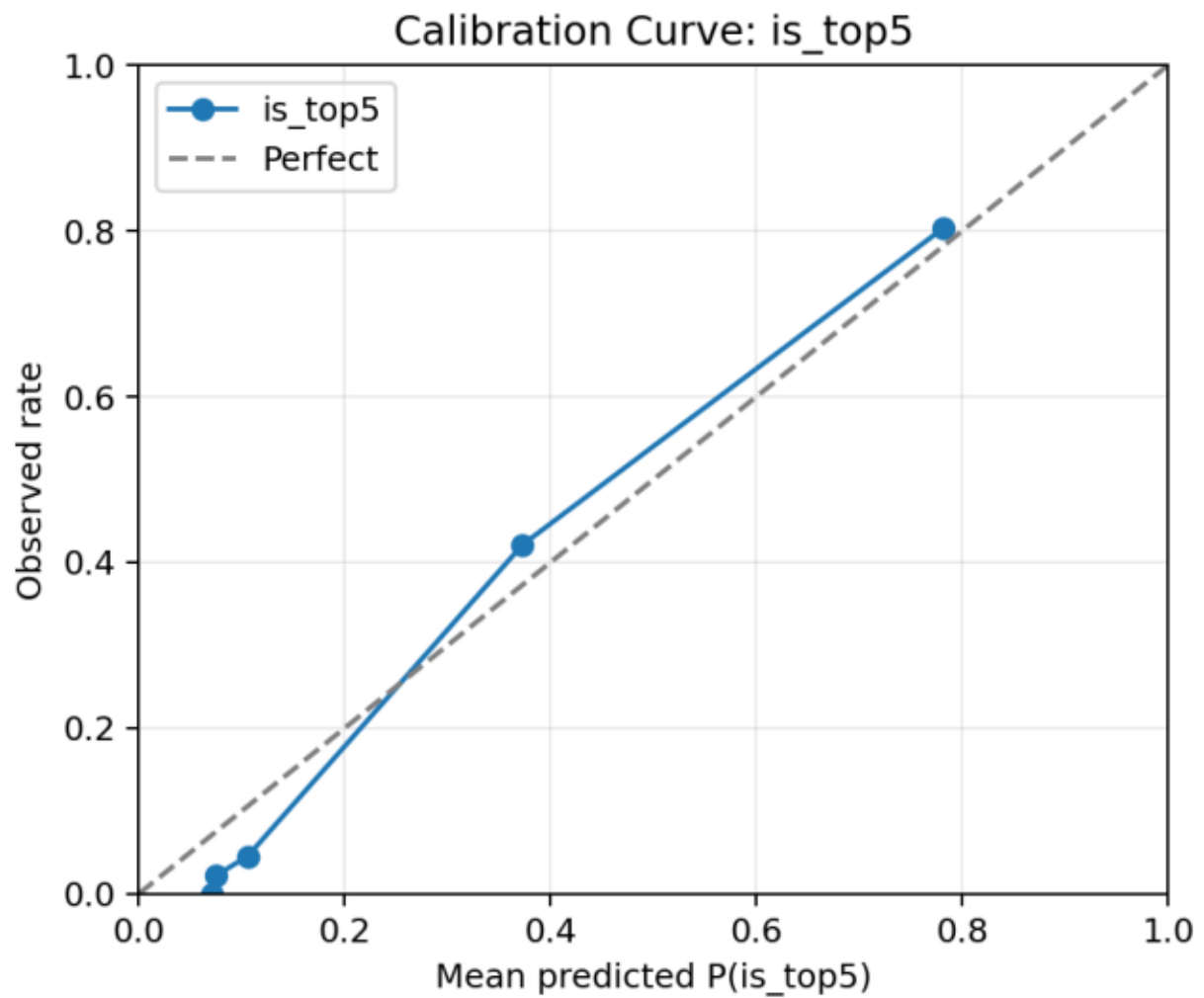
target	approach	brier_score	log_loss	roc_auc
is_top10	target_rate_baseline	0.250	0.693	0.500
is_top10	logistic_regression_calibra	0.139	0.443	0.881
is_top5	target_rate_baseline	0.192	0.572	0.500
is_top5	logistic_regression_calibra	0.090	0.306	0.934
is_top10	docent_reference	0.132	not provided	0.892

The Top 10 model improves strongly over the target-rate baseline but does not beat the docent floor. The Top 5 model has Brier 0.090479 and ROC-AUC 0.934413, which supports the expansion target as a useful decision lens.

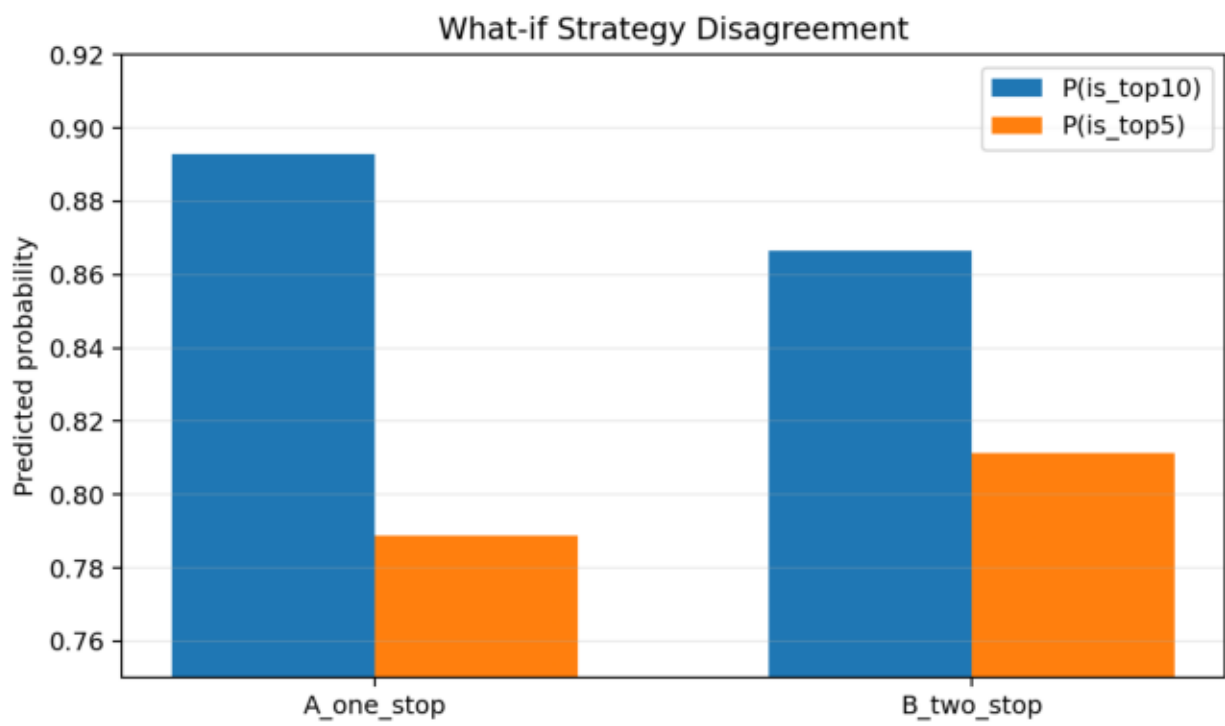
Calibration Evidence: is_top10



Calibration Evidence: is_top5



What-if Disagreement



6. Error Analysis and What-if

target	slice_column	slice_value	rows	actual_rate	predicted_prob	slice_brier_score
is_top5	strategy_type	three_plus_stop	153	0.196	0.225	0.103
is_top5	constructor_tier	front	170	0.676	0.646	0.181
is_top5	constructor_tier	midfield	407	0.278	0.287	0.114
is_top5	constructor_tier	backmarker	312	0.006	0.075	0.011
is_top10	strategy_type	three_plus_stop	153	0.405	0.482	0.157
is_top10	constructor_tier	midfield	407	0.619	0.625	0.159
is_top10	constructor_tier	backmarker	312	0.189	0.249	0.132
is_top10	constructor_tier	front	170	0.876	0.814	0.105

The main failure modes are midfield Top 10 predictions, three-plus-stop Top 10 strategies, and front-team Top 5 predictions. These are operational warnings, not reasons to hide the model.

7. Limitations and Risks

The strongest limitation is scenario dependence under regime shift and observational confounding. Strategy choice is not random, so the what-if output is not causal proof.

Single-team deployment is also risky because the selected scenario is midfield, and midfield Top 10 reliability is the weakest constructor-tier slice.

We do not recommend deploying this tool unless (1) the strategy desk limits it to pre-race what-if comparisons with fixed driver-race context, (2) every recommendation displays target-specific slice reliability for strategy type, circuit type, and constructor tier, and (3) the team validates the model on a fresh race weekend or simulator backtest.

8. Reproducibility and AI Reflection

Run `venv/bin/python Capstone/Final_Report/generate_final_report.py` from the repository root to regenerate the tables, figures, Markdown report, and PDF. The Hito 1 and Hito 2 source artifacts remain in `Capstone/` and all model `random_state` arguments use `RANDOM_SEED = 414`.

AI assistance was used for structure, prose, captions, critique, and Q&A drafting. Outputs were validated against the locked Hito evidence. Suggestions that overstated causality or claimed the Top 10 model beat the docent reference were rejected.

9. References

IIT414W course capstone brief and Canvas rubrics. (2026). F1 Race Strategy Advisor.

Course dataset: f1_strategy_race_level.csv, seasons 2019-2024.

FastF1 project documentation and historical Formula 1 timing data sources.

Jolpica F1 API documentation and historical race metadata sources.

scikit-learn developers. (2026). Logistic regression, calibration, and model evaluation documentation.